Journal of Travel Medicine, 2019, 1–2
doi: 10.1093/jtm/taz053
Editorial

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Pain control at the vaccine injection site: new insights

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Vaccination is a remarkable thing, a personal shield and a community shield against diseases that may otherwise claim millions of lives. Given that disaster is so routinely averted, the minor discomforts of the procedure—the needle pain and fear, the soreness or malaise—come much more into focus. Giving more attention to the side effects than the benefits of vaccination may lead individuals to refuse their 'shield' and diminish the community's protection, built by the combined contributions of vaccinated individuals that prevent transmission. Thus, herd immunity may become porous, leaving vulnerable those members who, for whatever reason, cannot receive their own vaccination shield.

International and national organizations have published position papers and clinical practice guidelines for pain mitigation during the vaccination procedure.[1,2] This is in recognition of the high proportion of parents who are concerned about vaccination pain in their children (24–40%) and who believe healthcare workers have a responsibility to make vaccinations less painful (80%).[3,4] Importantly, vaccination pain mitigation has been identified as an important strategy to counter vaccine hesitancy,[5] rated this year by the World Health Organization (WHO) as 1 of the 10 global health threats. General measures commonly recommended include specific guidelines for needle size/length and injection site selection, participant positioning and injection technique (no aspiration) and specific measures for infants and young children (caregiver presence/holding child, breastfeeding, distraction) and adults (breathing interventions).

However, a recent WHO position paper[1] highlights the need for research into specific interventions for adolescents and adults who can be used in mass comm nity campaigns and school-based programs. Here we highlight a novel approach to the issue of improving vaccine-related tolerance: the use of acute exercise to both reduce adverse events (including pain) and enhance the antibody response.

Exercise, the panacea for so many health issues, has recently been described as a potential contributor to better tolerating the vaccination process. Acute exercise has long been known to hold analgesic properties, with moderate to large effect sizes found to last for ~15 minutes after acute aerobic, isometric and dynamic resistance exercise.[6] Thus, it is remarkable that only very recently has an acute exercise intervention been used in the setting of vaccination to mitigate pain.

We recently examined the effect of exercise on vaccine-related pain, fear and anxiety in adolescents (aged 11–13 year) during vaccination against human papillomavirus (HPV). Using a 15-minute task of upper body exercises with simple resistance bands, we found that exercising prior to vaccine administration appeared to decrease vaccine-related pain in female adolescents, but curiously not in males who generally report less pain.[7] Sex differences were found in reported pain and anxiety in the control group (who received the vaccination according to usual-care procedures) but not the exercise group, suggesting that exercise diminished the negative experiences in the female adolescents to a lower level, equivalent to that reported by the male adolescents. This novel finding is, however, only one aspect of the argument for the benefit of acute exercise at the time of vaccination.

In the early 2000s, Dhabhar and others published a series of animal studies showing that acute stress could prime the immune system and enhance the response to a subsequent challenge.[8] Since then a number of human studies have been reported, examining the effects of exercise (both aerobic and resistance) on the immune response to vaccination. Although findings have been both positive (exercise groups showing stronger responses than control) and neutral (no difference observed between groups), the overall results suggest that exercise can enhance antibody and cell-mediated responses to vaccination, but these effects are more likely to be found against weaker immunogenic antigens.[9,10] Thus, in addition to pain mitigation, acute exercise may enhance the immune response to the vaccination.

Finally, recent data also suggest that acute exercise may reduce vaccine reactions, both local to the injection site and systemic, experienced after vaccinations.[11] In two randomised controlled trials (RCTs), one in adolescents receiving the HPV vaccine and one in young adults receiving the influenza vaccine, we found reductions in reported adverse reactions among both populations. Exercise significantly decreased reported days of tenderness in female adolescents compared to female controls. A trend towards a beneficial effect of exercise on reported days of feeling ill was also observed in adolescents after HPV vaccination. Furthermore, exercise had a beneficial effect in reducing swelling, fever and loss of appetite in young adults after influenza vaccination. The mechanism through which acute exercise may reduce adverse reactions is unknown but may include increases in lymph flow and lymphocytosis, which may together reduce the build-up of antigen around the site of injection. The reduction in systemic symptoms is not well understood, but similar findings have been found in other conditions: exercise has been reported to alleviate flu-like symptoms in multiple sclerosis patients receiving medications known to provoke flu-like symptoms.[12]

Given the WHO call for specific interventions in adolescents and adults, which can be used in mass community campaigns and school-based programs, the exercise tasks used in our recent studies deserve further practical evaluation. Most importantly, investigation is warranted of the cost and time associated with conducting the intervention, as well as the effectiveness of the intervention on vaccination pain and reactions. Our 15-minute task used resistance exercise bands, which ensured minimal cost associated with equipment. Furthermore, while machine-based resistance exercise requires individual supervision, exercise using resistance bands can be safely conducted in a group; particularly in a school setting. It seems reasonable that teachers would be well placed to conduct supervision. This has added benefit as exercising with another person or a group can improve motivation, increasing exercise performance and enjoyment.

Using a short bout of acute exercise offers a novel approach to mitigate pain at the injection site. The multi-aspect benefits of exercise in pain reduction, lessening of vaccine reactions and enhancement of immune responses against weaker antigens makes it an attractive intervention worthy of further research. It is even more attractive, given its easy implementation and that a short bout of exercise is inexpensive and requires little training to administer. It is well tolerated and familiar, avoiding potential public misunderstanding or fear, and it would be feasible for inclusion in mass community campaigns.

Author Contributions

K.E. and R.B. contributed to the idea conception, drafting and finalizing of the manuscript.

Conflict of Interest: None declared.

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